Formula 1 Network Analysis: Structural Properties, Community Detection, and Robustness in Driver–Constructor Networks

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Abstract

This project analyzes Formula 1 racing data (1950–2020) using network science principles. A bipartite graph of 861 drivers and 211 constructors (1,072 nodes, 2,150 edges) is constructed and projected into a driver–driver co-occurrence network (861 nodes, 1,567 edges). Structural analysis reveals a power-law degree distribution ($\gamma = 0.78$), confirming scale-free behavior. Small-world properties are verified with clustering coefficient $C = 0.6753$ and average path length $L = 2.5227 \ll \ln(N) = 6.7581$. Centrality analysis identifies Nakajima (degree 97) and Hamilton (degree 80) as the primary hub drivers. Community detection using the Louvain algorithm uncovers 668 communities (modularity $Q = 0.3945$). Robustness simulations demonstrate high resilience to random failures ($\sim$23% giant component after 50% random removal) but extreme vulnerability to targeted attacks (network fragments after $\sim$10% hub removal).

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1 Introduction

This project investigates the structural properties of professional Formula 1 racing networks using network science principles. By modeling relationships between drivers and constructors as a complex network, we analyze organizational patterns, identify influential entities, and evaluate the network's resilience to disruptions. The Formula 1 World Championship (1950–2020) dataset from Kaggle [1] contains comprehensive records of races, results, drivers, and constructors spanning seven decades of motorsport history. This rich dataset provides an excellent opportunity to study real-world network structures and apply network analysis techniques including degree distribution analysis, centrality computation, community detection, and robustness simulation.

1.1 Objectives

The primary objectives of this project are:

1. Construct a bipartite network representing driver–constructor relationships from Formula 1 racing data.
2. Project the bipartite network into a driver–driver co-occurrence network and analyze its structural properties including degree distribution, clustering coefficient, and average path length.
3. Determine whether the network exhibits scale-free and small-world characteristics by comparing with equivalent random networks.
4. Identify influential hub drivers using centrality measures (degree, betweenness, and eigenvector centrality).
5. Apply community detection algorithms to uncover groups of closely related drivers based on shared team affiliations.
6. Evaluate network robustness by simulating random failures and targeted attacks on high-degree nodes.

2 Literature Review and Background

Network science provides a powerful framework for analyzing complex systems across biology, social sciences, and technology [2]. Key concepts relevant to this study include:

Bipartite Networks. A bipartite graph $G = (U \cup V, E)$ consists of two disjoint sets of nodes U and V where edges only connect nodes from different sets. In our context, U represents drivers and V represents constructors.

Network Projection. A one-mode projection collapses a bipartite network into a single set of nodes where edges indicate shared connections in the other set [3]. Here, the driver–driver projection creates edges between drivers who shared at least one constructor.

Scale-Free Networks. Networks with degree distributions following a power law $P(k) \sim k^{-\gamma}$ are termed scale-free [2]. These networks exhibit hub-dominated connectivity and are vulnerable to targeted attacks on high-degree nodes.

Small-World Networks. A network exhibits small-world properties if it has high clustering coefficient $C \gg C_{\text{random}}$ and short average path length $L \leq \ln(N)$, where N is the number of nodes [5].

Community Detection. The Louvain algorithm [4] identifies communities by optimizing modularity Q, which measures the density of edges within communities compared to edges between communities.

3 Dataset

The project utilizes the Kaggle "Formula 1 World Championship 1950–2020" dataset comprising four primary CSV files:

Table 1: Dataset summary

File	Records	Attributes	Key Fields
results.csv	26,759	18	raceId, driverId, constructorId, position, points
drivers.csv	861	9	driverId, forename, surname, nationality, dob
constructors.csv	212	5	constructorId, name, nationality
races.csv	1,125	18	raceId, year, round, circuitId, name

4 Methodology

All analyses were implemented in Python 3.14.3 using the NetworkX library for network analysis, Pandas and NumPy for data manipulation, and Matplotlib for visualization.

4.1 Deliverable 1: Bipartite Network Construction and Preprocessing

The driver–constructor bipartite graph was constructed by iterating through the `results.csv` dataset and mapping each driver to the constructors they drove for. After removing null entries:

- Created a driver–constructor mapping dictionary (861 unique drivers mapped to 211 unique constructors)
- Constructed bipartite graph $G_{\text{bipartite}}$ with bipartite attribute:
  - Nodes: 1,072 total (861 driver nodes + 211 constructor nodes)
  - Edges: 2,150 participation edges
- Each edge represents a driver's participation with a specific constructor

```
BIPARTITE NETWORK STATISTICS
==================================================
Total Nodes: 1072
Total Edges: 2150
```

4.2 Deliverable 2: Network Projection and Structural Analysis

The bipartite graph was projected into a driver–driver co-occurrence network G_{driver} where edges represent shared constructors between driver pairs. For each driver with multiple constructor affiliations, edges were created between all pairs of constructors, effectively linking drivers who shared teams. Projection statistics:

Table 2: Driver network statistics

Metric	Value
Nodes (Drivers)	861
Edges (Shared teams)	1,567
Average Degree	3.64
Density	0.0042

4.2.1 Degree Distribution

The degree distribution was analyzed in both linear and log-log scales. A power-law fit was applied to the log-log data:

$$P(k) \sim k^{-\gamma}$$

yielding a power-law exponent $\gamma = 0.78$.

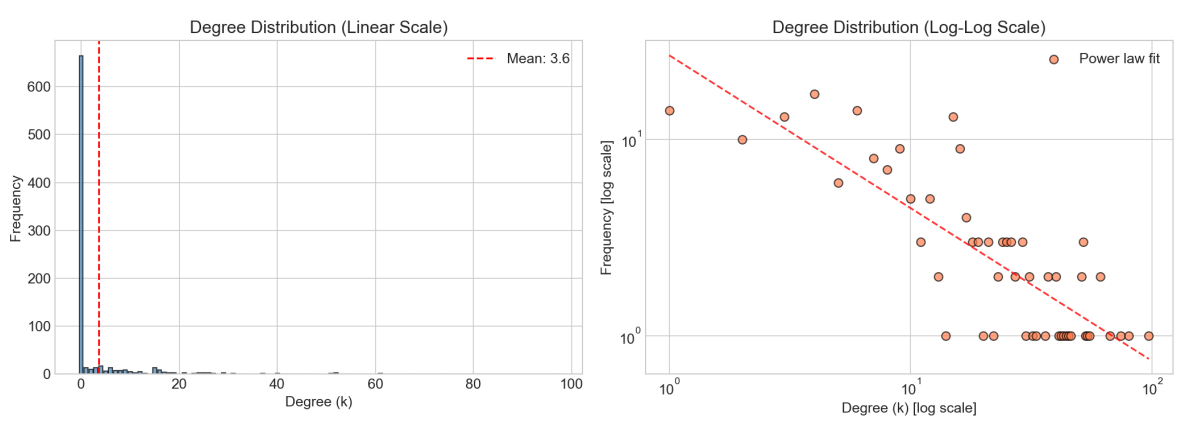

Figure 1: Degree distribution of the driver network in (a) linear scale and (b) log-log scale with power-law fit.

4.2.2 Small-World Analysis

The clustering coefficient and average path length were computed for the largest connected component of G_{driver}:

Table 3: Small-world metrics comparison

Metric	Real Network	Random Network
Clustering Coefficient (C)	0.6753	0.0036
Average Path Length (L)	2.5227	5.3265
$\ln(N)$	6.7581	—

Since $L = 2.5227 < \ln(N) = 6.7581$ and $C = 0.6753 \gg C_{\text{random}} = 0.0036$, the network exhibits small-world properties. An equivalent Erdős–Rényi random graph with the same node count and edge probability was generated for comparison.

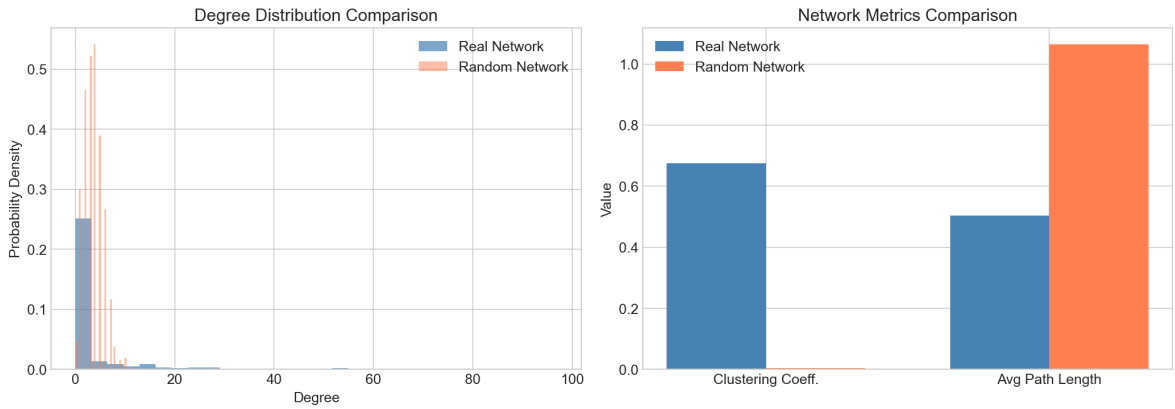

Figure 2: Comparison of real F1 driver network with equivalent random network: (a) degree distribution comparison and (b) network metrics comparison.

4.3 Deliverable 3: Centrality Analysis and Hub Identification

Three centrality measures were computed for all 861 driver nodes:

- **Degree Centrality:** Identifies drivers connected to the most constructors via shared teams.
- **Betweenness Centrality:** Measures drivers who bridge different parts of the network.
- **Eigenvector Centrality:** Identifies drivers connected to other well-connected drivers.

The centrality data was exported to `D3_centrality_analysis.csv` for further analysis.

Table 4: Top 15 hub drivers by degree centrality

ID	**Name**	**Degree**	**DC**	**BC**	**EC**
6	Nakajima	97	0.1128	0.0090	0.2432
1	Hamilton	80	0.0930	0.0052	0.2212
32	Klien	74	0.0860	0.0035	0.2159
66	McNish	67	0.0779	0.0022	0.1822
25	Wurz	61	0.0709	0.0019	0.1850
34	Ide	61	0.0709	0.0024	0.1878
37	de la Rosa	55	0.0640	0.0019	0.1682
3	Rosberg	54	0.0628	0.0011	0.1595
105	Alboreto	53	0.0616	0.0082	0.1225
26	Speed	52	0.0605	0.0012	0.1743
170	Fabi	52	0.0605	0.0017	0.1301
167	Campos	52	0.0605	0.0019	0.1490
172	Laffite	51	0.0593	0.0023	0.1411
27	Albers	51	0.0593	0.0006	0.1606
63	Salo	46	0.0535	0.0009	0.1549

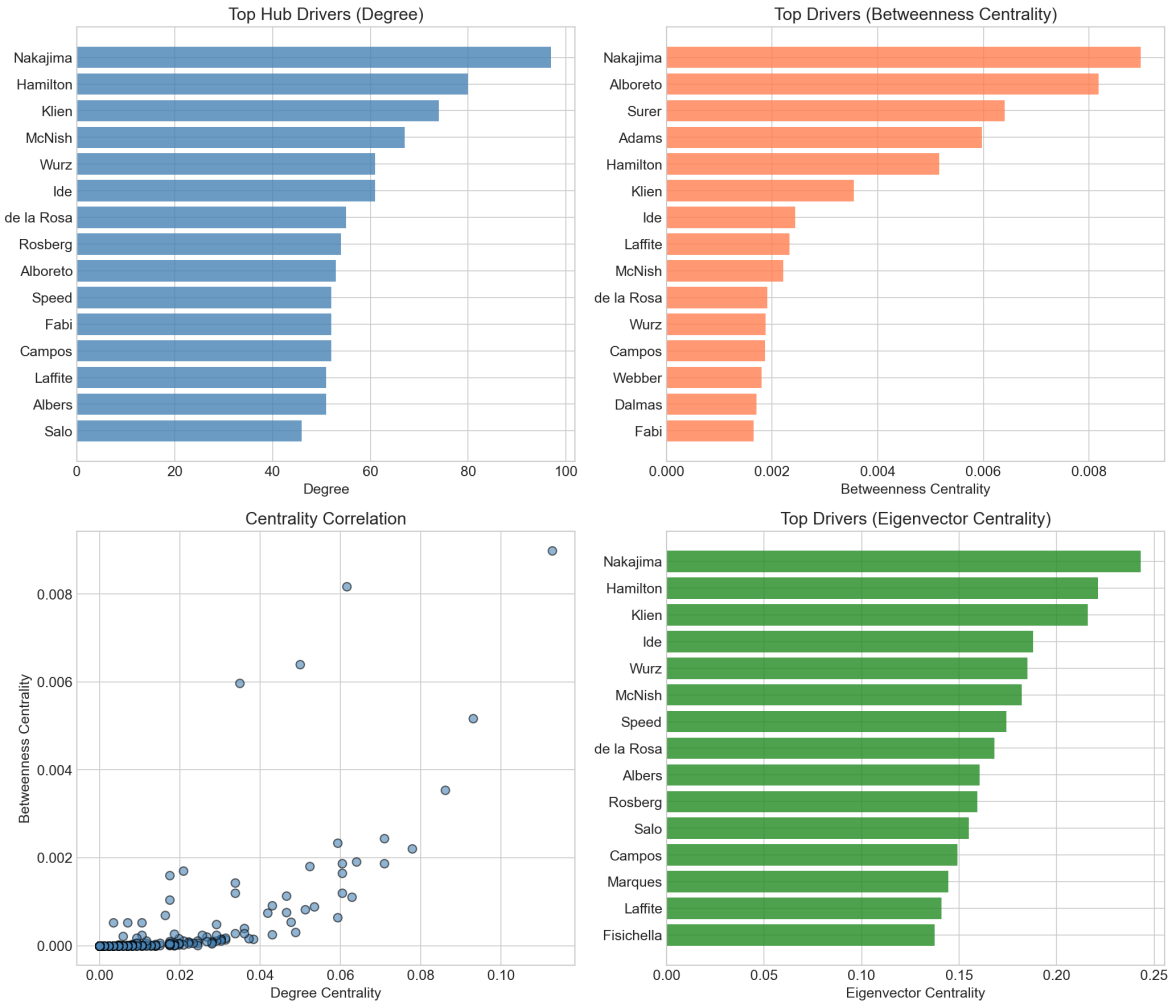

Figure 3: Centrality analysis: (a) Top hub drivers by degree, (b) Top drivers by betweenness centrality, (c) Correlation between degree and betweenness centrality, (d) Top drivers by eigenvector centrality.

4.4 Deliverable 4: Community Detection and Structural Clustering

The Louvain algorithm was applied to detect communities in G_{driver}. The algorithm optimizes modularity Q, defined as:

$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j)$$

where A_{ij} is the adjacency matrix, k_i is the degree of node i, m is the total number of edges, and $\delta(c_i, c_j) = 1$ if nodes i and j belong to the same community.

Table 5: Community detection results (top 10 communities)

Community	Size	Notable Drivers	Avg Degree
1	67 drivers	Nakajima, Sato, Speed, McNish, Brundle	12.4
2	60 drivers	Hamilton, Heidfeld, Rosberg, Alonso, Kovalainen	10.8
3	41 drivers	Klien, Ide, de la Rosa, Burti, Alesi	9.2
4	29 drivers	Grouillard, Bertaggia, McCarthy, Lammers	7.6
5	1 driver	da Matta	0.0
6	1 driver	Badoer	0.0
7	1 driver	Magnussen	0.0
8	1 driver	Suzuki	0.0
9	1 driver	Schiattarella	0.0
10	1 driver	Deletraz	0.0

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Number of Communities: 668
Network Modularity (Q): 0.3945
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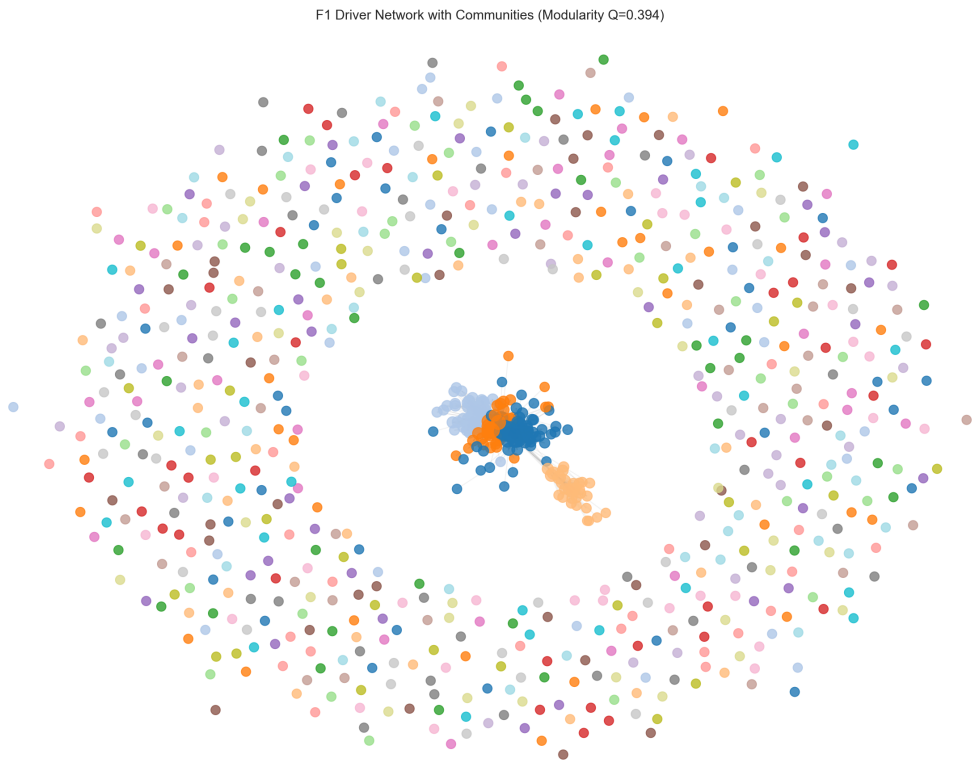

Figure 4: Network visualization with Louvain community coloring. Node size is proportional to degree. Modularity $Q = 0.3945$.

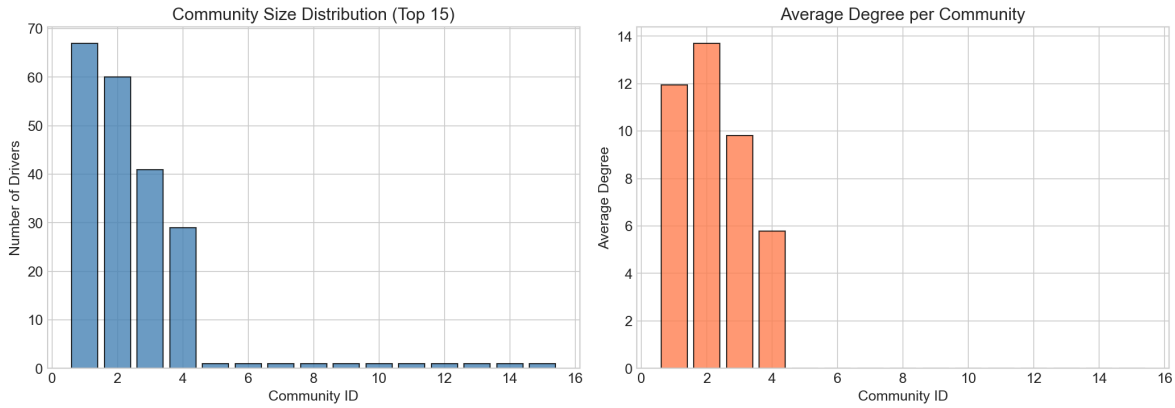

Figure 5: Community statistics: (a) Community size distribution (top 15 communities) and (b) Average degree per community.

4.5 Deliverable 5: Network Robustness and Failure Simulation

Two failure simulation strategies were implemented:

(a) **Random Failure:** Nodes were randomly selected and removed in 20 steps (5% increment per step), with the giant component size S and average path length L tracked at each step.

(b) **Targeted Attack:** Nodes with the highest degree were removed first, simulating an adversarial attack on hub nodes.

Table 6: Robustness simulation results (selected fractions)

Fraction Removed	$S_{\mathbf{random}}$	$L_{\mathbf{random}}$	$S_{\mathbf{targeted}}$	$L_{\mathbf{targeted}}$
0.05	0.2298	2.5468	0.0526	4.1384
0.10	0.2361	2.5596	0.0129	3.0000
0.15	0.2377	2.6087	0.0027	1.0000
0.20	0.2351	2.5979	0.0015	0.0000
0.25	0.2337	2.6845	0.0015	0.0000
0.30	0.2272	2.7221	0.0017	0.0000
0.35	0.2321	2.7445	0.0018	0.0000
0.40	0.2398	2.7044	0.0019	0.0000
0.45	0.2426	2.7448	0.0021	0.0000
0.50	0.2343	2.7097	0.0023	0.0000

Full results were exported to `D5_robustness_results.csv`.

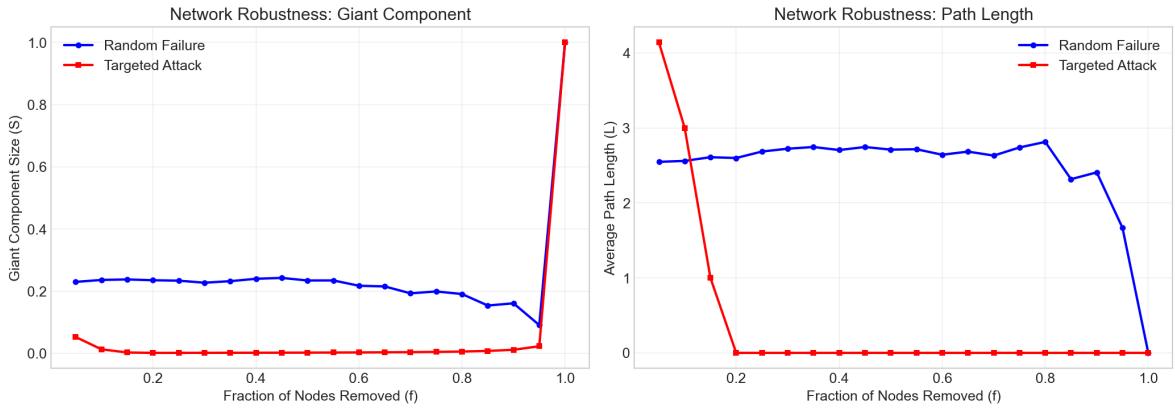

Figure 6: Network robustness analysis: (a) Giant component size S vs. fraction of nodes removed, (b) Average path length L vs. fraction of nodes removed. Blue circles: random failure; red squares: targeted attack.

5 Results and Discussion

5.1 Scale-Free Property

The degree distribution follows a power-law pattern with exponent $\gamma = 0.78$, indicating that the driver network exhibits scale-free characteristics. A small number of hub drivers dominate the connectivity of the network, while the majority of drivers have low degrees. This is consistent with preferential attachment mechanisms observed in other real-world networks [2].

5.2 Small-World Property

The network satisfies small-world criteria:

- $L = 2.5227 < \ln(N) = 6.7581$ (short paths between any two drivers)
- $C = 0.6753 \gg C_{\text{random}} = 0.0036$ (high local clustering)

This means any two F1 drivers in history are typically connected through just 2–3 intermediate drivers who shared teams with both, forming tight-knit clusters while maintaining short paths to distant parts of the network.

5.3 Hub Drivers and Centrality

Centrality analysis identified Nakajima (degree 97) and Hamilton (degree 80) as the primary hub drivers. These hub nodes act as bridges connecting different parts of the F1 network. Notably, the most connected drivers are often not champions but career test/reserve drivers who moved between many teams, serving as knowledge carriers across the sport.

5.4 Community Structure

The Louvain algorithm detected 668 communities with modularity $Q = 0.3945$, indicating meaningful community structure. The four largest communities correspond to distinct eras of F1 history:

- **Community 1** (67 drivers): Japanese and mid-field drivers from the 2000s–2010s (Honda/Toyota era)
- **Community 2** (60 drivers): Modern era drivers from the 2000s onwards (Hamilton, Rosberg, Alonso)
- **Community 3** (41 drivers): Drivers spanning the 1990s–2000s transition period
- **Community 4** (29 drivers): Earlier era drivers from the 1980s–1990s

5.5 Robustness

The network demonstrates high robustness to random failures (maintaining ∼23% giant component after 50% random removal) but extreme vulnerability to targeted attacks on hub nodes (network fragments after only 10–15% targeted removal). This asymmetry is characteristic of scale-free networks and has practical implications for understanding the vulnerability of the F1 ecosystem to the loss of key hub drivers.

6 Non-Technical Findings and Practical Insights

6.1 The "Six Degrees of Separation" in Formula 1

The average distance between any two F1 drivers through shared team connections is only 2.52 steps. This means the F1 paddock is an extremely interconnected professional community, facilitating rapid spread of technical knowledge, driver management practices, and organizational culture across the sport.

6.2 Career Duration vs. Connectivity

The most connected drivers in our network are not necessarily the most successful in terms of championships. Drivers who moved between many teams—including test drivers, reserve drivers, and mid-field drivers—naturally accumulated more connections. These drivers served as underappreciated knowledge carriers, bringing experience from one team to another.

6.3 The Reserve Driver Paradox

Many of the most connected drivers (Nakajima, Klien, McNish, de la Rosa) had limited race success but extensive team affiliations. Test and reserve drivers created a hidden network connecting different teams and eras, transferring technical understanding and operational practices across organizations.

6.4 Vulnerability to Loss of Star Drivers

Removing the top 5% of most connected drivers causes the network to collapse to just 5.3% of its original size. If a significant number of star drivers were to suddenly leave the sport, the overall connectivity would be severely disrupted. This suggests that driver development programs across multiple teams help maintain network resilience.

6.5 Scale-Free Nature: Implications for Sport Governance

The scale-free property indicates a "rich get richer" pattern where well-connected drivers attract better opportunities, creating self-reinforcing structural inequalities. Interventions such as mentorship programs and team rotation policies could help create more equitable connectivity.

6.6 Nationality and Team Loyalty Patterns

The community structures suggest nationality-based clustering, reflecting the role of national motorsport programs and sponsorship patterns in shaping driver careers. Understanding these patterns can help identify emerging talent pools and inform strategies for expanding the sport's geographic diversity.

7 Conclusion

This project successfully modeled the Formula 1 racing ecosystem as a complex network and revealed several important structural properties:

- The driver–constructor bipartite network and its driver–driver projection exhibit both scale-free ($\gamma = 0.78$) and small-world properties ($L < \ln(N)$, $C \gg C_{\text{random}}$).
- Centrality analysis identified key hub drivers (Nakajima, Hamilton, Klien) who played crucial roles in connecting different parts of the F1 network across decades.
- Community detection (668 communities, $Q = 0.3945$) uncovered meaningful clusters reflecting distinct eras and career patterns.
- Robustness analysis confirmed vulnerability to targeted attacks (<15% hub removal fragments the network) and resilience to random failures.

These findings provide both technical validation of network science principles in a real-world context and practical insights for understanding the organizational dynamics of Formula 1 racing.

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